COMPATIBILITY OF PERVERSE TRUNCATION OF NGÔ FIBRATIONS

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Abstract. We study compatibility of perverse truncations for morphisms between Ngô fibrations with rationally smooth total spaces. We show that, under an effective rank inequality, equivariant pullbacks preserve the normalized perverse truncations. Applied to the diagonal, this gives multiplicativity of perverse truncations for Hitchin systems with rationally smooth total spaces.

Contents

1. Introduction 1
1.1. Compatibility under equivariant comparison 2
1.2. Application to Hitchin fibrations: multiplicativity 3
1.3. Proof Strategy 4
1.4. Acknowledgements 4
2. Preliminaries 4
2.1. Comparisons and their induced morphisms 4
2.2. Perverse truncations and compatibility 5
2.3. Weak abelian fibrations 6
3. Compatibility of perverse truncation 7
3.1. Notation and the initial filtered comparison 8
3.2. Geometric bounds and the free exterior factor 12
3.3. From local degree bounds to global Ext vanishing 13
3.4. Obstruction to finer filtered morphisms 17
3.5. Conclusion of the main argument 18
4. Multiplicative perverse truncations for Hitchin Systems 19
4.1. The strongly parabolic Hitchin systems 19
4.2. Regular centralizers and Picard actions 21
4.3. Geometric hypotheses for the applications 22
4.4. Proof of multiplicativity 23
References 25

1. Introduction

Ngô introduced weak abelian fibrations and established his support theorem in the proof of the fundamental lemma [Ngô10, Section 7]. Commutative group actions and δ-regularity constrain the supports and local systems occurring in the decomposition theorem. De

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Cataldo–Rapagnetta–Saccà refined this description in terms of Ngô strings [dCRS21, Appendix A], while Maulik–Shen extended the support theorem to suitable equivariant complexes, including intersection complexes [MS23, Theorems 1.1 and 1.8]. More recently, de Cataldo–Fringuelli–Fernandez Herrero–Mauri introduced the notion of a Ngô fibration and proved that canonical and logarithmic Hitchin fibrations with positive twisting degree are Ngô fibrations [dCFFHM25, Definition 1.1 and Theorem A]. We follow the notion of Ngô fibration in [dCFFHM25, Definition 1.1]: a weak abelian fibration satisfying δ-regularity and a cohomological bound on the direct image of its intersection complex. For rationally smooth total spaces considered here, the cohomological bound follows from the fiber-dimension bound.

This paper studies compatibility of perverse truncations for Ngô fibrations with rationally smooth total spaces. We give a criterion for the natural pullback associated with an equivariant morphism of fibrations to preserve the perverse truncations. In fact, we study more general fibrations and the criterion is motivated by the δ-regularity and weak abelian fibrations.

Throughout, varieties and reductive groups are over $\mathbb{C}$. The basic object of study is a *comparison* (F, a), by which we mean a commutative square

$$\begin{array}{ccc} X & \xrightarrow{F} & Y \\ {\scriptstyle h_X}\downarrow & & \downarrow{\scriptstyle h_Y} \\ B_X & \xrightarrow[a]{} & B_Y. \end{array} \tag{1.1}$$

The square need not be Cartesian. Set

$$K_X = Rh_{X*}\mathbb{Q}_X, \qquad K_Y = Rh_{Y*}\mathbb{Q}_Y, \qquad d_X = \dim B_X, \quad d_Y = \dim B_Y.$$

Then we have the natural map:

$$\beta_{F,a} : a^*K_Y \longrightarrow K_X.$$

On both sides, we have the perverse truncations:

$${}^p\tau_{\leq d_X+k}K_X, \qquad {}^p\tau_{\leq d_Y+k}K_Y$$

Following conventions in [MSY26a], the indices are normalized by base dimension, so that for a proper fibration the index k agrees with fiber cohomological degree. We ask whether $\beta_{F,a}$ preserves the perverse truncations. More precisely, whether $\beta_{F,a}$ admits compatible factorizations

$$a^*({}^p\tau_{\leq d_Y+k}K_Y) \longrightarrow {}^p\tau_{\leq d_X+k}K_X, \qquad k \in \mathbb{Z}. \tag{1.2}$$

These factorizations imply

$$F^*P_kH^m(Y, \mathbb{Q}) \subseteq P_kH^m(X, \mathbb{Q}).$$

We call (1.2) *the compatibility of perverse truncation* with respect to the comparison.

1.1. **Compatibility under equivariant comparison.** Let (X, P_X, B_X) and (Y, P_Y, B_Y) be projective weak abelian fibrations, as in [MS23, Section 1.1], see also Section 2.3. Assume that the bases are smooth, the commutative group schemes are smooth separated with connected fibers, and, for $Z = X, Y$,

$$\dim Z = 2d_Z, \qquad \text{every nonempty fiber of } h_Z \text{ is pure of dimension } d_Z.$$

Assume also that X and Y are rationally smooth, in particular their intersection complexes are shifted constant sheaves.

Suppose the comparison (F, a) is equivariant for a smooth separated finite-type commutative group algebraic space Q/B_X with connected fibers and homomorphisms

$$P_X \xleftarrow{u} Q \xrightarrow{v} a^*P_Y.$$

Explicitly, $F_a : X \to Y \times_{B_Y} B_X$, $x \mapsto (F(x), h_X(x))$, satisfies $F_a(u(q) \cdot x) = v(q) \cdot F_a(x)$. Write $\mathrm{Ab}(P_b)$ for the abelian quotient of a connected commutative algebraic group P_b, and set

$$r(b) = \dim \operatorname{im}\big(\mathrm{Ab}(Q_b) \longrightarrow \mathrm{Ab}(P_{X,b})\big), \qquad \dim\{b \in B_X : r(b) \le c\} \le c \quad (c \ge 0). \tag{1.3}$$

The bound implies the δ-regularity of (X, P_X, B_X). If the map on abelian quotients is surjective, the bound is precisely δ-regularity of P_X/B_X.

Theorem 1.1. *Under these hypotheses, the comparison (F, a) is compatible with perverse truncation in the sense of Definition 2.2. In particular, F^* preserves the perverse filtration.*

This is the weak-abelian-fibration case of Theorem 3.5. If the rank bound is relaxed to $\dim\{r \le c\} \le c + \kappa$, the conclusion has target ${}^{p}\tau_{\le d_X+k+\kappa} K_X$.

Remark 1.2. Theorem 3.5 requires (X, P_X, B_X) to be a weak abelian fibration. The right hand side needs a smooth separated commutative group action, but neither polarizability nor affine stabilizers. The weak-abelian-fibration formulation above is a convenient special case.

1.2. **Application to Hitchin fibrations: multiplicativity.** Hitchin fibrations provide the main applications. Their perverse filtrations enter the $P = W$ conjecture of de Cataldo–Hausel–Migliorini [dCHM12], which, under non-abelian Hodge theory, identifies the perverse filtration P on the Dolbeault moduli space M_{Dol} with the weight filtration W on the corresponding twisted character variety M_{B}:

$$P_k H^m(M_{\mathrm{Dol}}, \mathbb{Q}) = W_{2k} H^m(M_{\mathrm{B}}, \mathbb{Q}) = W_{2k+1} H^m(M_{\mathrm{B}}, \mathbb{Q}).$$

For curves of genus at least two, the coprime GL_n conjecture was proved by Maulik–Shen [MS24] and Hausel–Mellit–Minets–Schiffmann [HMMS25]; Maulik–Shen–Yin gave a Fourier-theoretic proof [MSY25, Theorem 0.4]. Maulik–Shen–Yin also established motivic multiplicativity for dualizable abelian fibrations satisfying Fourier vanishing, whose realization gives multiplicativity of perverse truncations [MSY25, Theorem 2.6(ii)–(iii)]; see also [MSY26b, Theorem 3.2(b)].

The multiplicativity of perverse truncations plays an important role in the proof of the $P = W$ conjecture. To see how the compatibility can be used to prove multiplicativity: the diagonal morphism is equivariant, and its pullback is cup-product; hence multiplicativity follows from the compatibility.

Let (X, P, B) be a Ngô fibration with $d = \dim B$. The comparison

$$(\Delta_X, \Delta_B) : (X \to B) \longrightarrow (X \times X \to B \times B)$$

is equivariant for P acting diagonally on the product. The Künneth formula identifies the "$\beta_{F,a}$" with the cup product.

Theorem 1.3. *For $K = Rh_*\mathbb{Q}_X$ and $r, s \in \mathbb{Z}$, cup-product factors as*

$$ {}^{p}\tau_{\leq r}K \otimes^{\mathbf{L}} {}^{p}\tau_{\leq s}K \longrightarrow {}^{p}\tau_{\leq r+s-d}K \longrightarrow K. \tag{1.4}$$

Consequently $P_iH^m(X,\mathbb{Q}) \cup P_jH^n(X,\mathbb{Q}) \subseteq P_{i+j}H^{m+n}(X,\mathbb{Q})$. This includes the classical GL_N Higgs moduli spaces, the twisted traceless SL_N spaces $M^{\Lambda,0}_{\mathrm{SL}_N}(C,\varnothing)$, and the full-flag strongly parabolic spaces, with the specified determinant and weights where applicable, under Assumption 4.2.

The shift is determined by the comparison of bases: $\dim(B\times B) = 2d$, whereas $\dim B = d$. Substituting $r = d + i$ and $s = d + j$ gives the normalized bound $d + i + j$.

1.3. **Proof Strategy.** The argument uses the weak abelian fibration structure, rational smoothness and the effective-rank inequality (1.3). For a Ngô fibration, the last condition follows from δ-regularity when the common group surjects onto the abelian quotients of the source symmetry groups. For the application to Hitchin systems, the geometric hypotheses are collected in Assumption 4.2.

The proof first uses the group action to make $\beta_{F,a}$ a morphism of modules of the homological algebras. The abelian part of the action on X then supplies a free exterior-algebra factor. Combined with stalk and costalk bounds, this gives a global Ext vanishing range controlled by (1.3) which removes the obstruction to the compatible factorization (1.2).

Section 2 introduces comparison morphisms, perverse truncations and weak abelian fibrations. Section 3 proves the compatibility criterion and its extensions. Section 4 gives the applications: deriving multiplicativity from the diagonal.

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2. Preliminaries

Cohomology has rational coefficients if not specified. We write $D^b_c(B,\mathbb{Q})$ for the bounded constructible derived category, and ${}^{p}\tau$ and ${}^{p}\mathcal{H}$ for middle-perversity truncation and cohomology. The comparison morphisms and compatibility with perverse truncations are initially considered in the ordinary constructible derived categories. The enhanced categories used in the proof are specified in Subsection 3.1.

2.1. Comparisons and their induced morphisms.

Definition 2.1. A *comparison* from $h_X : X \to B_X$ to $h_Y : Y \to B_Y$ is a pair of algebraic morphisms

$$F : X \to Y, \qquad a : B_X \to B_Y, \qquad h_Y \circ F = a \circ h_X.$$

We assume the morphisms h_X, h_Y are projective and the bases are smooth. Set $Y_a = Y \times_{B_Y} B_X$, let $h_{Y,a} : Y_a \to B_X$ be its projection, and define $F_a : X \to Y_a$ by $F_a(x) = (F(x), h_X(x))$. We have the following diagram:

$$\begin{array}{ccccc} X & \xrightarrow{F_a} & Y_a & \longrightarrow & Y \\ & \searrow^{h_X} & \downarrow{\scriptstyle h_{Y_a}} & & \downarrow{\scriptstyle h_Y} \\ & & B_X & \xrightarrow{a} & B_Y \end{array}$$

where the right square is Cartesian. Proper base change gives the natural map

$$\beta_{F,a} : a^* K_Y \xrightarrow{\sim} Rh_{Y,a*}\mathbb{Q}_{Y_a} \longrightarrow Rh_{Y,a*}RF_{a*}\mathbb{Q}_X = K_X. \tag{2.1}$$

In other words, β is the pushforward of the natural map $\mathbb{Q}_{Y_a} \to RF_{a*}\mathbb{Q}_X$ by Rh_{Y_a*}.

2.2. **Perverse truncations and compatibility.** Put $d_X = \dim B_X$ and $d_Y = \dim B_Y$. We use the base-dimension normalization customary for the Hitchin perverse filtration, as in [MSY26a, Section 1.1, (2)–(4)], and define

$$P_k H^m(X, \mathbb{Q}) = \operatorname{im}\big(H^m(B_X, {}^p\tau_{\le d_X+k} K_X) \longrightarrow H^m(X, \mathbb{Q})\big). \tag{2.2}$$

For a smooth proper map, ${}^p\mathcal{H}^{d_X+q}(K_X) = (R^q h_{X*}\mathbb{Q}_X)[d_X]$, which explains the normalization. Equivalently, if $\widehat{K}_X = K_X[d_X]$, then

$$P_k H^m(X, \mathbb{Q}) = \operatorname{im}\big(H^{m-d_X}(B_X, {}^p\tau_{\le k}\widehat{K}_X) \longrightarrow H^m(X, \mathbb{Q})\big).$$

Maulik–Shen–Yin[MSY26a] shift their direct image by $\dim X - d_h$, where d_h is the relative dimension. In our dimension range $\dim X = 2d_X$ and $d_h = d_X$, this is exactly $\widehat{K}_X$. In particular the absolute truncation bound in their Conjecture 6.1 is $r + s - (\dim X - d_h) = r + s - d_X$, as in (1.4). A convention using $Rh_{X*}\,\mathrm{IC}_X = K_X[2d_X]$ instead has the centered perverse index $k - d_X$; it is not the normalization used here. Arbitrary inverse image does not commute with perverse truncation, so it is important to retain the order of operations in the following definition.

Definition 2.2. The comparison (F, a) is *compatible with perverse truncation* if for any $k \in \mathbb{Z}$, the composite has the factorization

$$\begin{array}{ccccc} a^{*p}\tau_{\le d_Y+k} K_Y & \longrightarrow & a^* K_Y & \xrightarrow{\beta_{F,a}} & K_X \\ & \searrow & & \nearrow & \\ & & {}^p\tau_{\le d_X+k} K_X & & \end{array} \tag{2.3}$$

The comparison morphisms and compatibility with perverse truncations are formulated in the ordinary bounded constructible derived category.

Remark 2.3. If h_X is projective and X is a rational homology manifold, the decomposition theorem implies that each canonical morphism ${}^p\tau_{\le m} K_X \to K_X$ is a split monomorphism in the derived category. Consequently, a factorization through this morphism, when it exists, is unique. The factorizations in (2.3) therefore commute automatically with the truncation transition maps.

Applying hypercohomology to (2.3), immediately gives

$$F^* P_k H^m(Y, \mathbb{Q}) \subseteq P_k H^m(X, \mathbb{Q}).$$

For comparisons $(F, a) : (X \to B_X) \to (Y \to B_Y)$ and $(G, b) : (Y \to B_Y) \to (Z \to B_Z)$, we have

$$\beta_{G\circ F, b\circ a} = \beta_{F,a} \circ a^* \beta_{G,b}.$$

If both comparisons are compatible, their compatible factorizations compose as

$$(b \circ a)^{*p}\tau_{\leq d_Z+k} K_Z = a^* b^{*p}\tau_{\leq d_Z+k} K_Z \longrightarrow a^{*p}\tau_{\leq d_Y+k} K_Y \longrightarrow {}^p\tau_{\leq d_X+k} K_X. \tag{2.4}$$

Thus compatibility of perverse truncation is preserved by composition, with no additional normalized shift.

2.3. **Weak abelian fibrations.** In general, the compatibility of perverse truncation cannot be expected. We here introduce the *weak abelian fibrations* which play an important role in the study of Hitchin systems. We follow [MS23, Section 1.1]. Let $h : M \to B$ be proper, with M quasiprojective, and let $g : P \to B$ be a smooth commutative group scheme with geometrically connected fibers, acting on M over B. All the group schemes used here are of finite type.

Definition 2.4. The triple (M, P, B) is a *weak abelian fibration of relative dimension* s if

(1) every fiber of $P \to B$ has dimension s, and M is pure-dimensional with $\dim M = s + \dim B$;
(2) the action of P on M has affine stabilizers;
(3) the relative Tate module of P/B is polarizable.

The definition in [dCFFHM25, Definition 4.36] permits disconnected group fibers and imposes polarizability on the neutral component. Passing to that component gives the connected-fiber convention used here.

More precisely, the Tate module can be written

$$T_{\overline{\mathbb{Q}}_\ell}(P) = \mathcal{H}^{-1}\big(Rg_! \overline{\mathbb{Q}}_\ell[2s](s)\big).$$

If we consider the rational coefficients, its stalk is $H_1(P_b, \mathbb{Q})$. Following [MS23, Section 1.3], polarizability means that an *étale-local* bilinear pairing descends, on each fiber, to a non-degenerate pairing on the Tate module of the abelian quotient. Equivalently, its fiberwise radical is the Tate module of the connected affine part.

For a point b, by Chevalley's theorem,

$$1 \longrightarrow P_b^{\mathrm{aff}} \longrightarrow P_b \longrightarrow \mathrm{Ab}(P_b) \longrightarrow 1, \qquad \delta(b) = \dim P_b^{\mathrm{aff}}.$$

The group scheme is δ-*regular* if

$$\mathrm{codim}_B\{b : \delta(b) \geq m\} \geq m \qquad (m \geq 0). \tag{2.5}$$

When $s = \dim B = d$, the identity $\dim \mathrm{Ab}(P_b) = d - \delta(b)$ rewrites this as

$$\dim\{b : \dim \mathrm{Ab}(P_b) \leq c\} \leq c \qquad (c \geq 0).$$

Definition 2.4 fixes $\dim M = s + \dim B$ and already requires h to be proper. The applications impose the further conditions $s = \dim B$, smoothness of B, rational smoothness of M, projectivity of h, pure fiber dimension s, and δ-regularity. These extra conditions are stated separately whenever they are needed.

A pure-dimensional variety M is rationally smooth, or a rational homology manifold, when $\mathbb{Q}_M[\dim M] \cong \mathrm{IC}_M$. This allows us to apply the decomposition theorem to the constant direct images carrying the natural pullback.

3. Compatibility of perverse truncation

Motivated by *weak abelian fibrations*, we consider comparisons with a common action by a commutative group space.

Definition 3.1. Let $(F, a) : (X \to B_X) \to (Y \to B_Y)$ be a comparison. We say it is an *equivariant comparison*, if additionally, there is a smooth separated finite-type commutative group algebraic space Q/B_X with connected fibers and homomorphisms $u : Q \to P_X$ and $v : Q \to a^*P_Y$ such that:

$$F_a\big(u(q) \cdot x\big) = v(q) \cdot F_a(x).$$

Throughout this section, we make the following geometric assumptions on the comparisons

Assumption 3.2. *For notational convenience, we denote $Z = X, Y$.*

1. *Z is a pure-dimensional rational homology manifold (RHM).*
2. *$d_Z = \dim B_Z, \dim Z = 2d_Z, \dim h_Z^{-1}(b) = d_Z$, for every nonempty fiber, which is required to be pure-dimensional.*
3. *We have P_Z/B_Z, smooth separated finite-type commutative group scheme with connected fibers, acting on Z.*
4. *(X, P_X, B_X) is a weak abelian fibration, i.e., the relative Tate module $T_{\bar{\mathbb{Q}}_\ell}(P_X)$ is étale-locally polarizable and the P_X-stabilizers are affine.*

We then need the following numerical assumption on *the equivariant comparison.*

Assumption 3.3. *Put*

$$r(b) = \dim \mathrm{im}(\mathrm{Ab}\, Q_b \longrightarrow \mathrm{Ab}\, P_{X,b}).$$

We call the following:

$$\dim\{b \in B_X : r(b) \le c\} \le c \quad (c \ge 0)$$

effective-rank inequality.

Remark 3.4. Suppose P_X/B_X has relative dimension d_X, and put $A(b) = \dim \mathrm{Ab}(P_{X,b})$. Since $r(b) \le A(b)$, the effective-rank inequality implies

$$\dim\{b : A(b) \le c\} \le \dim\{b : r(b) \le c\} \le c,$$

which is precisely δ-regularity of P_X/B_X.

Theorem 3.5. *Under these hypotheses, the comparison (F, a) is compatible with perverse truncation. More precisely, for every $k \in \mathbb{Z}$, the natural composite*

$$a^*({}^p\tau_{\le d_Y+k} K_Y) \longrightarrow a^*K_Y \xrightarrow{\beta_{F,a}} K_X$$

factors as

$$a^*({}^p\tau_{\le d_Y+k} K_Y) \longrightarrow {}^p\tau_{\le d_X+k} K_X \longrightarrow K_X. \tag{3.1}$$

Consequently, $F^*P_kH^m(Y,\mathbb{Q}) \subseteq P_kH^m(X,\mathbb{Q})$.

The proof is organized into four parts. The geometric input is isolated in Proposition 3.11; the numerical calculation is contained in Lemma 3.13 and Proposition 3.14. The final two lemmas improve the filtered degree and descend the resulting truncation factorizations to rational coefficients. The proof of Theorem 3.5 appears after these results.

3.1. **Notation and the initial filtered comparison.** Throughout the proof we retain the geometric hypotheses of Theorem 3.5. The intermediate constructions have coefficients in $\mathbb{Q}_\ell$; compatibility over $\mathbb{Q}$ is obtained in Lemma 3.16.

For each base B, let $\mathscr{D}(B)$ denote the ambient presentable symmetric monoidal stable ∞-category in the pro-étale realization of sheaf complexes used below. Its tensor product preserves colimits separately in each variable. We use the enhanced six operations of [LZ24] and the realization of bounded constructible complexes in [BS15, Lemma 6.5.5 and Proposition 6.8.14]. Writing $\mathscr{D}^b_c(B)$ for the bounded constructible subcategory, we have

$$h\mathscr{D}^b_c(B) \simeq D^b_c(B,\mathbb{Q}_\ell),$$

where h denotes the homotopy category.

Algebras are unital associative algebra objects in $\mathscr{D}(B)$, and module structures and module morphisms are understood in this enhanced setting. For an algebra object $\mathcal{A}$, we write

$$D(\mathcal{A}) = h\,\mathrm{Mod}_{\mathcal{A}}(\mathscr{D}(B)).$$

The ambient categories accommodate module constructions, direct sums, and Rees objects; bounded constructibility is imposed on the geometric complexes under consideration.

Compatibility in Definition 2.2 is a property in the ordinary constructible derived category. The enhanced module structures introduced here are used to prove that property.

For a smooth morphism f of relative dimension q, denote its smooth trace map by

$$\mathrm{Tr}_f : Rf_!\mathbb{Q}_\ell[2q](q) \longrightarrow \mathbb{Q}_\ell.$$

Let $\pi : Q \to B_X$ have relative dimension q, and set $\mathcal{A} = R\pi_!\mathbb{Q}_\ell[2q](q)$. Consider the following Cartesian diagram:

$$\begin{array}{ccc} Q \times_{B_X} Q & \xrightarrow{p} & Q \\ \downarrow{\scriptstyle m} & & \downarrow{\scriptstyle \pi} \\ Q & \xrightarrow{\pi} & B_X \end{array}$$

By the base change for $R\pi_!$, we have:

$$\pi^*R\pi_!\mathbb{Q}_\ell[2q](q) \simeq Rm_!\mathbb{Q}_\ell[2q](q).$$

Hence, the group multiplication $m : Q \times_{B_X} Q \to Q$ induces the multiplication on $\mathcal{A}$:

$$\mathcal{A}\otimes\mathcal{A} \simeq R\pi_!(\mathbb{Q}_\ell[2q](q)\otimes\pi^*R\pi_!\mathbb{Q}_\ell[2q](q)) \simeq R\pi_!Rm_!\mathbb{Q}_\ell[4q](2q) \xrightarrow{R\pi_!(\mathrm{Tr}_m)[2q](q)} \mathcal{A},$$

where the first map is the Künneth isomorphism. The unit of the algebra is induced by the identity section $e : B_X \to Q$.

The homomorphisms $u : Q \to P_X$ and $v : Q \to a^*P_Y$ induce actions of Q on X and Y_a, respectively. For $(Z,h_Z) = (X,h_X)$ or (Y_a,h_{Y_a}), write $\alpha_Z : Q \times_{B_X} Z \to Z$ for this action and $p_Z : Q \times_{B_X} Z \to Z$ for projection.

The isomorphism $(g, z) \mapsto (g, gz)$ identifies α_Z with p_Z, so it is smooth of relative dimension q and we have the following Cartesian diagram:

$$\begin{array}{ccc} Q \times_{B_X} Z & \xrightarrow{p} & Q \\ \downarrow{\scriptstyle \alpha_Z} & & \downarrow{\scriptstyle \pi} \\ Z & \xrightarrow{h_Z} & B_X \end{array}$$

Similarly, the action by Q gives a natural module structure:
(3.2)

$$c_Z : \mathcal{A} \otimes Rh_{Z*}\mathbb{Q}_{\ell,Z} \simeq Rh_{Z!}(h_Z^*\mathcal{A} \otimes \mathbb{Q}_{\ell,Z}) \simeq Rh_{Z!}R\alpha_{Z!}\mathbb{Q}_{\ell,Q\times_{B_X}Z}[2q](q) \xrightarrow{Rh_{Z!}(\mathrm{Tr}_{\alpha_Z})} Rh_{Z*}\mathbb{Q}_{\ell,Z}.$$

These are the cap-product actions on K_X and a^*K_Y.

Lemma 3.6. *These operations make $\mathcal{A}$ a bounded constructible algebra and a^*K_Y, K_X into $\mathcal{A}$-modules. The natural morphism $\beta : a^*K_Y \to K_X$ is a morphism of these $\mathcal{A}$-modules.*

Proof. The six-functor formalism of [LZ24, Theorems 7.2.6–7.2.7] applies to the separated algebraic space Q. Since π is of finite presentation, [LZ24, Proposition 7.4.1] implies that $\mathcal{A}$ is constructible. For a geometric point b of B_X, the base-change theorem for $R\pi_!$ identifies $\mathcal{A}_b$ with $R\Gamma_c(Q_b, \mathbb{Q}_\ell)[2q](q)$ [LZ24, Theorem 7.2.7(2)]. Since Q_b has dimension q, its compactly supported cohomology vanishes outside $[0, 2q]$. Thus $\mathcal{A}$ is concentrated in degrees $[-2q, 0]$.

Since we use the cap-product construction of [Ngô10, Sections 7.4.2–7.4.3] here, its associativity for equivariant coefficients is proved in [MS23, Proposition 1.4]. It remains to show that the natural morphism $\beta : a^*K_Y \to K_X$ respects these module structures. In other words, we need to show the commutativity of the following diagram:

$$\begin{array}{ccccc} \mathcal{A} \otimes a^*K_Y & \xrightarrow{\simeq} & \mathcal{A} \otimes Rh_{Y_a*}\mathbb{Q}_{\ell,Y_a} & \xrightarrow{1\otimes\beta} & \mathcal{A} \otimes K_X \\ \downarrow & & \downarrow{\scriptstyle c_{Y_a}} & & \downarrow{\scriptstyle c_X} \\ a^*K_Y & \xrightarrow{\simeq} & K_{Y_a} := Rh_{Y_a*}\mathbb{Q}_{\ell,Y_a} & \xrightarrow{\beta} & K_X \end{array}$$

By the definition of the module structure, c_Z can be decomposed as follows, see (3.2), we have:

$$\begin{array}{ccc} \mathcal{A} \otimes Rh_{Y_a*}\mathbb{Q}_{\ell,Y_a} & \xrightarrow{1\otimes\beta} & \mathcal{A} \otimes K_X \\ \downarrow{\scriptstyle \simeq} & & \downarrow{\scriptstyle \simeq} \\ Rh_{Y_a*}(h_{Y_a}^*R\pi_!\mathbb{Q}_{\ell,Q}[2q](q)) & & Rh_{X*}(h_X^*R\pi_!\mathbb{Q}_{\ell,Q}[2q](q)) \\ \downarrow{\scriptstyle \simeq} & & \downarrow{\scriptstyle \simeq} \\ Rh_{Y_a*}(R\alpha_{Y_a!}\mathbb{Q}_\ell[2q](q)) & \dashrightarrow & Rh_{X*}(R\alpha_{X!}\mathbb{Q}_\ell[2q](q)) \\ \downarrow{\scriptstyle Rh_{Y_a*}\,\mathrm{Tr}} & & \downarrow{\scriptstyle Rh_{X*}\,\mathrm{Tr}} \\ K_{Y_a} := Rh_{Y_a*}\mathbb{Q}_{\ell,Y_a} & \xrightarrow{\beta} & K_X \end{array}$$

Hence, it suffices to show the commutativity of the lowest square which follows from the functoriality of the trace map. More precisely, apply the base-change isomorphism to the

Cartesian square of action morphisms:

$$\begin{array}{ccc} Q \times_{B_X} X & \longrightarrow & Q \times_{B_X} Y_a \\ \downarrow{\scriptstyle \alpha_X} & & \downarrow{\scriptstyle \alpha_{Y_a}} \\ X & \xrightarrow{F_a} & Y_a \end{array}$$

we obtain

$$F_a^* R\alpha_{Y_a!}\mathbb{Q}_\ell[2q](q) \simeq R\alpha_{X!}\mathbb{Q}_\ell[2q](q),$$

compatibility of smooth trace with base change gives $F_a^*(\mathrm{Tr}_{\alpha_{Y_a}}) = \mathrm{Tr}_{\alpha_X}$; see [LZ24, Remark 4.1.6(2), Theorem 7.2.6].

The Q-equivariance of F_a gives

$$F_a((g_1 \cdots g_n)x) = (g_1 \cdots g_n)F_a(x) \qquad (n \geq 1).$$

These identities are compatible with multiplying adjacent g_i and inserting the identity of Q. Applying the enhanced trace and base-change formalism of [LZ24, Remark 4.1.6, Theorems 7.2.6–7.2.7] therefore makes β a morphism of $\mathcal{A}$-modules in the stable ∞-category. □

Lemma 3.7. *The homology algebra of Lemma 3.6 admits a natural equivalence of algebra objects*

$$\mathcal{A} \simeq \bigoplus_{k \geq 0} \Lambda^k T(Q)[k]$$

in the symmetric monoidal stable ∞-category, functorial in Q.

Proof. Commutativity of Q and the enhanced Künneth and trace formalism equip $\mathcal{A}$ with a commutative algebra structure; see [LZ24, Theorems 7.2.6–7.2.7]. Fix $N \geq 2$ and write $\psi = [N]_*$. By [Ngô10, Sections 7.4.3–7.4.4], $\mathcal{H}^{-k}(\mathcal{A}) = \Lambda^k T(Q)$ as a graded algebra, with ψ acting by N^k.

For $m \geq \max\{1, 2q\}$ and $0 \leq j \leq m$, put

$$p_j(t) = \prod_{\substack{0 \leq k \leq m \\ k \neq j}} (t - N^k), \qquad \mathcal{A}_j = \mathrm{colim}\big(\mathcal{A} \xrightarrow{p_j(\psi)} \mathcal{A} \xrightarrow{p_j(\psi)} \cdots\big),$$

where the colimit is taken in the derived ∞-category of sheaves of $\mathbb{Q}_\ell$-vector spaces on the pro-étale site of B_X. Since filtered colimits are exact and $p_j(N^k)$ vanishes precisely for $k \neq j$, the canonical maps give

$$\mathcal{A} \xrightarrow{\sim} \bigoplus_{j=0}^{m} \mathcal{A}_j, \qquad \mathcal{A}_j \simeq \mathcal{H}^{-j}(\mathcal{A})[j].$$

Thus the $j = 1$ summand defines $T(Q)[1] \to \mathcal{A}$, inducing the identity on $\mathcal{H}^{-1}$. Group homomorphisms commute with ψ, and enlarging m only introduces factors invertible on the surviving cohomology. Hence this inclusion is functorial in Q.

By the universal property of the free commutative algebra, the inclusion extends to

$$\mathrm{Sym}(T(Q)[1]) \simeq \bigoplus_{k \geq 0} \Lambda^k T(Q)[k] \longrightarrow \mathcal{A};$$

see [Lur17, Proposition 3.1.3.13 and Example 3.1.3.14]. Indeed, permutations of the shifted factors act on $T(Q)^{\otimes k}[k]$ with the sign character. Since $k!$ is invertible in $\mathbb{Q}_\ell$, this gives $\mathrm{Sym}^k(T(Q)[1]) \simeq \Lambda^k T(Q)[k]$. The resulting algebra morphism induces the exterior-product isomorphism on every cohomology sheaf, hence is an equivalence, functorially in Q. □

Lemma 3.6 gives $[\beta] \in \mathrm{Hom}_{D(\mathcal{A})}(a^*K_Y, K_X)$. Put $F_n^Y K_Y = a^*({}^p\tau_{\le n}K_Y)$ and $F_n^X K_X = {}^p\tau_{\le n}K_X$. We regard filtrations as objects of

$$\mathrm{Seq}(\mathscr{D}(B_X)) = \mathrm{Fun}((\mathbb{Z}, \le), \mathscr{D}(B_X)),$$

equipped with the Day-convolution tensor product. The underlying object of a sequence is its colimit.

Put $F_\ell\mathcal{A} = \tau^{\mathrm{ord}}_{\le \ell}\mathcal{A}$. This is an algebra object in $\mathrm{Seq}(\mathscr{D}(B_X))$. By a filtered $\mathcal{A}$-module we mean a module over $F_\bullet\mathcal{A}$ in this category.

A filtered $\mathcal{A}$-linear lift of the fixed enhanced comparison β of degree s is a morphism

$$\widetilde{\beta}_s : F_\bullet^Y K_Y \longrightarrow F_{\bullet+s}^X K_X$$

of filtered $\mathcal{A}$-modules, together with an identification of its colimit with the given $\mathcal{A}$-module morphism β.

All the truncation filtrations used here vanish in sufficiently negative degrees, and their colimits recover the corresponding unfiltered objects.

Lemma 3.8. *The cap-product actions equip $F_\bullet^Y K_Y$ and $F_\bullet^X K_X$ with filtered $\mathcal{A}$-module structures. The comparison β admits a filtered $\mathcal{A}$-linear lift of degree d_X:*

$$\widetilde{\beta}_{d_X} : F_\bullet^Y K_Y \longrightarrow F_{\bullet+d_X}^X K_X.$$

Proof. Write $\mathcal{A}_{P_Z}$ for the homology algebra of P_Z/B_Z, with its ordinary truncation filtration. Tensoring by an ordinary sheaf preserves ${}^pD^{\le n}$, so

$$F_\ell\mathcal{A}_{P_Z} \otimes {}^p\tau_{\le n}K_Z \in {}^pD^{\le n+\ell}(B_Z), \qquad Z = X, Y.$$

The cap-product map therefore factors through ${}^p\tau_{\le n+\ell}K_Z$. Restrict the X-action along $Q \to P_X$; for Y, first pull back from B_Y and then restrict along $Q \to a^*P_Y$.

Also,

$$a^*({}^p\tau_{\le n}K_Y) \in D^{\mathrm{ord},\le n}(B_X) \subset {}^pD^{\le n+d_X}(B_X),$$

so the composite with β factors through ${}^p\tau_{\le n+d_X}K_X$.

These factorizations retain the enhanced module structures. Indeed, truncation adjunction gives

$$\mathrm{Map}_{\mathscr{D}(B)}(C, {}^p\tau_{\le m}K) \simeq \mathrm{Map}_{\mathscr{D}(B)}(C, K) \qquad (C \in {}^pD^{\le m}).$$

The degree bounds above apply to all iterated actions, so the actions and the comparison factor through the truncations together with their coherences. For sufficiently large n, the truncations are the full complexes and the resulting comparison is β. □

Remark 3.9. The lemma gives an initial truncation bound. To obtain compatibility in (2.3), we will improve the degree from d_X to $d_X - d_Y$ while retaining the filtered $\mathcal{A}$-module structure. This improvement is obtained by the vanishing of the corresponding obstruction groups.

3.2. **Geometric bounds and the free exterior factor.** Set

$$\mathcal{P}_i = {}^{p}\mathcal{H}^{d_Y+i}K_Y, \qquad \mathcal{R}_j = {}^{p}\mathcal{H}^{d_X+j}K_X, \qquad \Delta = d_Y - d_X.$$

Choose a finite stratification of B_X by smooth locally closed subvarieties S such that $r(b)$ is constant on each S, and the restrictions of the homology sheaves of Q and the sheaves

$$U_i^u = \mathcal{H}^u(\iota^* a^* \mathcal{P}_i), \qquad V_j^v = \mathcal{H}^v(\iota^! \mathcal{R}_j), \qquad \iota : S \hookrightarrow B_X,$$

are local systems on S for all i, j, u, v. Write $t = \dim S$. Here i, j are perverse indices, and u, v are ordinary cohomological degrees.

Lemma 3.10. *On every smooth stratum S of dimension t, the ordinary coefficient systems satisfy*

$$U_i^u \neq 0 \Rightarrow d_Y + u \leq i \leq d_Y - u, \qquad V_j^v \neq 0 \Rightarrow d_X - v - 2t \leq j \leq d_X + v + 2t. \tag{3.3}$$

Proof. Fix $b \in S$. Since $\mathcal{P}_i[-d_Y - i]$ is a direct summand of K_Y, proper base change identifies $(U_i^u)_b$ with a direct summand of $H^{d_Y+i+u}(Y_{a(b)}, \mathbb{Q}_\ell)$. The fiber dimension bound therefore gives $i + u \leq d_Y$ whenever $(U_i^u)_b \neq 0$. Relative hard Lefschetz gives $\mathcal{P}_i \simeq \mathcal{P}_{2d_Y - i}(d_Y - i)$. Applying the same bound yields $2d_Y - i + u \leq d_Y$, hence $i \geq d_Y + u$.

Since $\mathbb{D}K_X \simeq K_X[4d_X](2d_X)$, we have $\mathbb{D}\mathcal{R}_j \simeq \mathcal{R}_{2d_X - j}(2d_X)$. Verdier duality on the smooth stratum S, where the cohomology sheaves are locally constant, gives

$$(V_j^v)_b \simeq \left(\mathcal{H}^{-v-2t}(\iota^* \mathcal{R}_{2d_X - j})_b\right)^\vee (t - 2d_X).$$

Thus $(V_j^v)_b \neq 0$ implies $2d_X - j \geq d_X - v - 2t$ by the stalk bound for X, equivalently $j \leq d_X + v + 2t$. Apply the relative hard Lefschetz again, $\mathcal{R}_j \simeq \mathcal{R}_{2d_X - j}(d_X - j)$, then we have $j \geq d_X - v - 2t$. □

For $b \in S$, let

$$f_b : T(Q)_b \xrightarrow{T(u)_b} T(P_X)_b \longrightarrow T(\operatorname{Ab} P_{X,b}), \qquad r = \frac{1}{2} \dim_{\mathbb{Q}_\ell} \operatorname{im} f_b.$$

Fix a subspace $A_Q \subset T(Q)_b$ such that $f_b|_{A_Q} : A_Q \xrightarrow{\sim} \operatorname{im} f_b$.

Proposition 3.11. *Let $A_Q \subset T(Q)_b$ as above with $\dim A_Q = 2r$. For every ordinary cohomological degree v, the graded module $\bigoplus_j (V_j^v)_b$ is free over ΛA_Q under the action induced by Lemma 3.6.*

Proof. Since S is smooth of dimension t, by Verdier duality,

$$\mathcal{H}^v(\iota^! \mathcal{R}_j)_b \simeq \left(\mathcal{H}^{-v-2t}(\iota^* \mathcal{R}_{2d_X - j})_b\right)^\vee (t - 2d_X).$$

Fix a lift $\lambda : T(\operatorname{Ab} P_{X,b}) \to T(P_{X,b})$ and put $A = \operatorname{im} \lambda$. Each vector $\xi \in A$ is a generator of ΛA in grading degree -1. The costalk operator

$$\xi : \mathcal{H}^v(\iota^! \mathcal{R}_j)_b \longrightarrow \mathcal{H}^v(\iota^! \mathcal{R}_{j-1})_b$$

corresponds under Verdier duality to the transpose of

$$\xi : \mathcal{H}^{-v-2t}(\iota^* \mathcal{R}_{2d_X - j + 1})_b \longrightarrow \mathcal{H}^{-v-2t}(\iota^* \mathcal{R}_{2d_X - j})_b,$$

with the usual grading signs and the sign $\xi \mapsto -\xi$ induced by group inversion. Thus the costalk module carries the graded dual action of the stalk module.

By [Ngô10, Propositions 7.4.10, 7.5.5, 7.7.4 and Lemma 7.7.5] (see also [MS23, Proposition 1.5 and Section 1.6, Step B]), $\bigoplus_j \mathcal{H}^m(\iota^*\mathcal{R}_j)_b$ is a free ΛA-module. Writing $m = \dim A$, the perfect pairing $\Lambda^k A \otimes \Lambda^{m-k} A \to \Lambda^m A$ shows that the graded dual of a finite free ΛA-module is free. Twisting the action by $\xi \mapsto -\xi$ also preserves freeness. Hence $\bigoplus_j \mathcal{H}^v(\iota^!\mathcal{R}_j)_b$ is free over ΛA.

Recall that $u : Q \to P_X$ is the homomorphism in our assumptions. Since $T(u)_b(A_Q)$ projects injectively to $T(\mathrm{Ab}\, P_{X,b})$, we may take the subspace $A \subset T(P_X)_b$ in the preceding argument to satisfy

$$T(u)_b(A_Q) \subset A, \qquad A \xrightarrow{\sim} T(\mathrm{Ab}\, P_{X,b}).$$

Naturality of the cap product action identifies this restriction with the action induced by Q. Since ΛA is free over ΛA_Q, the freeness then follows. □

Remark 3.12. Proposition 3.11 also holds without the RHM assumption if K_X is replaced by $Rh_{X*}\mathrm{IC}_X[-\dim X]$. Indeed, the strict-support freeness also holds for equivariant IC coefficients [MS23, Propositions 1.4–1.6]. The arguments for the costalk and the restriction along $Q \to P_X$ are unchanged.

3.3. **From local degree bounds to global Ext vanishing.** Put

$$\mathrm{Gr}(\mathscr{D}(B_X)) = \mathrm{Fun}(\mathbb{Z}_{\mathrm{disc}}, \mathscr{D}(B_X)),$$

with its Day-convolution tensor product. For an algebra object $\mathcal{B}$ in this category, define

$$D_{\mathrm{gr}}(\mathcal{B}) = h\,\mathrm{Mod}_{\mathcal{B}}\big(\mathrm{Gr}(\mathscr{D}(B_X))\big).$$

The associated graded construction is given by

$$(\mathrm{gr}\, F)_n = \mathrm{cofib}(F_{n-1} \to F_n).$$

It is symmetric monoidal for the relevant convolution products; see [GP18, Section 2.23 and Proposition 2.26]. It therefore sends filtered algebras and modules to graded algebras and modules.

When $\mathcal{B}$ is concentrated in cohomological degree zero, $D_{\mathrm{gr}}(\mathcal{B})$ identifies with the derived category of sheaves of internally graded $\mathcal{B}$-modules. For a graded complex C, use the shearing convention

$$(\mathsf{S}C)_n = C_n[n],$$

with the usual Koszul signs.

Since

$$\mathrm{gr}^F_{-k}\,\mathcal{A} \simeq \Lambda^k T(Q)[k],$$

the sheared associated graded algebra has cohomology concentrated in degree zero and is quasi-isomorphic to

$$E = \Lambda T(Q), \qquad E_{-k} = \Lambda^k T(Q).$$

Shearing and reindexing the associated graded modules give objects $U, V \in D_{\mathrm{gr}}(E)$ with components

$$U_i = (\mathrm{gr}^F_{d_Y+i}\, F^Y K_Y)[d_Y + i] \simeq a^*\mathcal{P}_i,$$
$$V_j = (\mathrm{gr}^F_{d_X+j}\, F^X K_X)[d_X + j] \simeq \mathcal{R}_j.$$

with the E-module structures induced by the filtered cap-product actions. In particular, their stalk and costalk cohomology modules are precisely the U_i^u and V_j^v considered above.

Write $R\mathcal{H}om_B^{\mathrm{gr}}(L,M)$ for the internal derived Hom of graded B-modules, preserving internal degree, and put $\mathcal{E}xt_B^{e,\mathrm{gr}}(L,M)=\mathcal{H}^eR\mathcal{H}om_B^{\mathrm{gr}}(L,M)$. Over a base W, global derived Hom is

$$R\mathrm{Hom}_B^{\mathrm{gr}}(L,M)=R\Gamma\big(W,R\mathcal{H}om_B^{\mathrm{gr}}(L,M)\big).$$

In particular,

$$\mathrm{Hom}_{D_{\mathrm{gr}}(\mathcal{B})}(L,M[e])=H^eR\Gamma\big(B_X,R\mathcal{H}om_{\mathcal{B}}^{\mathrm{gr}}(L,M)\big).$$

The superscript gr requires preservation of internal degree. The shifts $[e]$ and $\langle\delta\rangle$ refer to cohomological and internal degree, respectively. Finally, define

$$(V\langle\delta\rangle)_i=V_{i+\delta},\qquad \mathrm{Ext}^e_{E,\delta}(U,V)=\mathrm{Hom}_{D_{\mathrm{gr}}(E)}\big(U,V\langle\delta\rangle[e]\big).$$

Lemma 3.13. *On a pro-étale cover of a stratum S of dimension t trivializing $T(Q)|_S$ and the graded local systems $U^u=\bigoplus_i U_i^u$ and $V^v=\bigoplus_j V_j^v$, a nonzero morphism $f:U^u\to V^v\langle\lambda\rangle$ of graded ΛA_Q-modules satisfies*

$$\lambda\le-\Delta+v-u+2(t-r). \tag{3.4}$$

Here $\Delta=d_Y-d_X$, $\dim A_Q=2r$.

Proof. On such a pro-étale cover of S, split $T(Q)=A_Q\oplus R_0$ as vector spaces. Let $f:U^u\to V^v\langle\lambda\rangle$ be a nonzero morphism of graded ΛA_Q-modules, with components $f_i:U_i^u\to V_{i+\lambda}^v$. Since the augmentation ideal of ΛA_Q is nilpotent, there exist an index i and an element $x\in U_i^u$ such that

$$0\neq y:=f_i(x)\in V_{i+\lambda}^v,\qquad \xi y=0\quad\text{for every }\xi\in A_Q.$$

We now bound the degrees of x and y. Choose a basis $a_1,\dots,a_{2r}$ of A_Q. Each a_k acts by maps $V_j^v\to V_{j-1}^v$. By Proposition 3.11, choose free generators $e_\alpha\in V_{h_\alpha}^v$ such that

$$V^v=\bigoplus_\alpha(\Lambda A_Q)e_\alpha.$$

Since $e_\alpha\neq0$, Lemma 3.10 gives $h_\alpha\le d_X+v+2t$.

In ΛA_Q, an element is annihilated by every a_k if and only if it is a scalar multiple of $a_1\cdots a_{2r}$. Consequently, the relations $a_ky=0$ and freeness imply $y=\sum_\alpha c_\alpha\cdot(a_1\cdots a_{2r})e_\alpha$. Each nonzero term belongs to $V_{h_\alpha-2r}^v$. Since $0\neq y\in V_{i+\lambda}^v$, at least one c_α is nonzero, and for every such α,

$$i+\lambda=h_\alpha-2r\le d_X+v+2t-2r.$$

On the other hand, $f_i(x)=y\neq0$ implies $U_i^u\neq0$, so Lemma 3.10 gives $i\ge d_Y+u$. Combining these inequalities yields

$$\lambda\le(d_X+v+2t-2r)-(d_Y+u)=-\Delta+v-u+2(t-r),$$

which proves (3.4).

□

Proposition 3.14. *For the objects $U,V\in D_{\mathrm{gr}}(E)$ defined above, the effective-rank inequality 3.3, i.e., $r\ge t$, on every stratum implies*

$$\mathrm{Ext}^e_{E,\delta}(U,V)=0\qquad\text{if }\ e<\Delta+\delta.$$

More generally, if $r\ge t-\kappa$ on every stratum, the vanishing range is $e<\Delta+\delta-2\kappa$.

Proof. Fix a stratum S, passing to the pro-étale covering of S if necessary, we may assume $T(Q) = A_Q \oplus R_0$. Then $E = \Lambda A_Q \otimes_{\mathbb{Q}_\ell} \Lambda R_0$, and the tensor product is graded: for $\xi \in R_0$ and $a \in \Lambda A_Q$ of degree $|a|$, $\xi a = (-1)^{|a|} a\xi$. All Hom and Ext operations in the following local calculation are internal and preserve the grading.

For each fixed v, Proposition 3.11 makes V^v a free graded ΛA_Q-module. For fixed u, v, Lemma 3.13 gives

$$\mathcal{H}om^{\mathrm{gr}}_{\Lambda A_Q}\big(U^u, V^v\langle\lambda\rangle\big) = 0 \qquad \text{if } \lambda > -\Delta + v - u + 2(t-r).$$

After refining the pro-étale cover to trivialize $T(Q)$, U^u and V^v, internal Ext is computed by sheafifying resolutions of ΛA_Q-modules by finite free graded modules. Since ΛA_Q is a finite-dimensional Frobenius algebra, every finite free graded ΛA_Q-module is injective in the category of graded modules. Thus

$$\mathcal{E}xt^{q,\mathrm{gr}}_{\Lambda A_Q}\big(U^u, V^v\langle\lambda\rangle\big) = 0 \qquad (q > 0).$$

Define a graded sheaf $H = \bigoplus_\lambda H_\lambda$ by

$$H_\lambda = \mathcal{H}om^{\mathrm{gr}}_{\Lambda A_Q}\big(U^u, V^v\langle\lambda\rangle\big).$$

Thus a local section $f \in H_\lambda$ is a family of maps $f_i : U^u_i \longrightarrow V^v_{i+\lambda}$ satisfying $f(am) = af(m)$.

For $\xi \in R_0$, define its action on H by

$$(\xi \star f)(m) = (-1)^i\big(\xi_V f(m) - f(\xi_U m)\big), \qquad m \in U^u_i,$$

which defines a ΛA_Q-linear map. Here the subscripts in ξ_U, ξ_V indicate the module on which ξ acts. The operators satisfy $\xi\star(\eta\star f)+\eta\star(\xi\star f) = 0$, hence make H a graded ΛR_0-module.

The condition $\xi \star f = 0$ for every $\xi \in R_0$ is equivalent to $\xi_V f(m) = f(\xi_U m)$. Together with ΛA_Q-linearity, this is precisely E-linearity. Consequently,

$$\mathcal{H}om^{\mathrm{gr}}_E\big(U^u, V^v\langle\delta\rangle\big) \simeq \mathcal{H}om^{\mathrm{gr}}_{\Lambda R_0}\big(\mathbb{Q}_\ell, H\langle\delta\rangle\big),$$

where $\mathbb{Q}_\ell$ is concentrated in degree zero with trivial R_0-action.

We have an identity of composite functors:

$$\mathcal{H}om^{\mathrm{gr}}_E(U^u, -) \simeq \mathcal{H}om^{\mathrm{gr}}_{\Lambda R_0}(\mathbb{Q}_\ell, -) \circ \left(\bigoplus_{\lambda\in\mathbb{Z}} \mathcal{H}om^{\mathrm{gr}}_{\Lambda A_Q}\big(U^u, (-)\langle\lambda\rangle\big)\right).$$

The inner functor carries the ΛR_0-action defined above. It is right adjoint to the exact functor $W \longmapsto U^u \otimes_{\mathbb{Q}_\ell} W$, with E-action

$$a(m\otimes w) = am \otimes w, \qquad \xi(m\otimes w) = \xi_U m \otimes w + (-1)^i m \otimes \xi_W w, \qquad m \in U^u_i.$$

It therefore sends injective graded E-modules to injective graded ΛR_0-modules. Since E is free over ΛA_Q, restriction to ΛA_Q also preserves injective modules. Hence we have the Grothendieck spectral sequence

$$E_2^{p,q} = \mathcal{E}xt^{p,\mathrm{gr}}_{\Lambda R_0}\left(\mathbb{Q}_\ell, \left[\bigoplus_{\lambda\in\mathbb{Z}} \mathcal{E}xt^{q,\mathrm{gr}}_{\Lambda A_Q}\big(U^u, V^v\langle\lambda\rangle\big)\right]\langle\delta\rangle\right)$$
$$\Longrightarrow \mathcal{E}xt^{p+q,\mathrm{gr}}_E\big(U^u, V^v\langle\delta\rangle\big).$$

By the internal Ext vanishing above, all rows with $q > 0$ vanish. Thus

$$\mathcal{E}xt^{n,\mathrm{gr}}_E\big(U^u, V^v\langle\delta\rangle\big) \simeq \mathcal{E}xt^{n,\mathrm{gr}}_{\Lambda R_0}\big(\mathbb{Q}_\ell, H\langle\delta\rangle\big) \qquad (n \geq 0).$$

Thus these Ext sheaves are computed by applying $\mathcal{H}om^{\mathrm{gr}}_{\Lambda R_0}(-, H\langle\delta\rangle)$ to the Koszul resolution of $\mathbb{Q}_\ell$.

The nth term of the Koszul resolution is $P_n = \Lambda R_0 \otimes_{\mathbb{Q}_\ell} \mathrm{Sym}^n R_0$ with free generators in degree $-n$. A degree-preserving map $P_n \to H\langle\delta\rangle$ sends these generators to $H_{\delta-n}$, whose elements are ΛA_Q-linear maps $U_i^u \longrightarrow V_{i+\delta-n}^v$. By Lemma 3.13,

$$\mathcal{H}om^{\mathrm{gr}}_{\Lambda R_0}\big(P_n, H\langle\delta\rangle\big) = 0, \qquad \text{if } \ \delta - n > -\Delta + v - u + 2(t-r).$$

Taking cohomology and using the preceding Ext isomorphism gives

$$\mathcal{E}xt_E^{n,\mathrm{gr}}\big(U^u, V^v\langle\delta\rangle\big) = 0 \qquad \text{if } \ n < \Delta + \delta + u - v + 2(r-t)$$

We reduce the global calculation to the strata. Let

$$\iota : Z \hookrightarrow B_X, \qquad j : O = B_X \setminus Z \hookrightarrow B_X$$

be a closed inclusion and its open complement. The E-action on V induces an ι^*E-action on $\iota^!V$. In fact,

$$\iota_*(\iota^*E \otimes \iota^!V) \simeq E \otimes \iota_*\iota^!V \longrightarrow E \otimes V \longrightarrow V.$$

Pushing forward this action gives an E-module structure on $\iota_*\iota^!V$ through

$$E \otimes \iota_*\iota^!V \simeq \iota_*(\iota^*E \otimes \iota^!V) \longrightarrow \iota_*\iota^!V.$$

Similarly, Rj_*j^*V is also an E-module.

Thus the localization triangle $\iota_*\iota^!V \longrightarrow V \longrightarrow Rj_*j^*V \xrightarrow{+1}$ is a triangle of E-modules. Applying $R\mathrm{Hom}^{\mathrm{gr}}_E(U, -)$ and using adjunction gives a triangle

$$R\mathrm{Hom}^{\mathrm{gr}}_{\iota^*E}(\iota^*U, \iota^!V) \longrightarrow R\mathrm{Hom}^{\mathrm{gr}}_E(U, V) \longrightarrow R\mathrm{Hom}^{\mathrm{gr}}_{j^*E}(j^*U, j^*V) \xrightarrow{+1}.$$

Induction over strata reduces the desired vanishing to the terms

$$R\mathrm{Hom}^{\mathrm{gr}}_{E_S}(i_S^*U, i_S^!V), \qquad i_S : S \hookrightarrow B_X, \quad E_S = i_S^*E.$$

We already get the vanishing $\mathcal{E}xt^{n,\mathrm{gr}}_{E_S}(U^u, V^v\langle\delta\rangle) = 0$ if $n < \Delta + \delta + u - v + 2(r-t)$. Since $r \geq t$, the vanishing implies that for all pairs (u, v),

$$\mathcal{H}^e R\mathcal{H}om^{\mathrm{gr}}_{E_S}\big(U^u[-u], V^v[-v]\langle\delta\rangle\big) = \mathcal{E}xt^{e+u-v,\mathrm{gr}}_{E_S}\big(U^u, V^v\langle\delta\rangle\big) = 0 \qquad \text{if } \ e < \Delta + \delta.$$

The E-module truncation triangles

$$\tau_{\leq v-1}(i_S^!V) \longrightarrow \tau_{\leq v}(i_S^!V) \longrightarrow V^v[-v] \xrightarrow{+1}$$

and the corresponding triangles for i_S^*U are finite. Applying internal derived Hom and inducting in both variables therefore gives

$$R\mathcal{H}om^{\mathrm{gr}}_{E_S}\big(i_S^*U, i_S^!V\langle\delta\rangle\big) \in D^{\geq\Delta+\delta}.$$

Taking derived global sections preserves this lower bound. The module localization triangles then give

$$\mathrm{Ext}^e_{E,\delta}(U, V) = 0 \qquad \text{if } \ e < \Delta + \delta$$

on the whole base.

If $r \geq t - \kappa$, the same argument gives

$$\mathrm{Ext}^e_{E,\delta}(U, V) = 0 \qquad \text{if } \ e < \Delta + \delta - 2\kappa.$$

□

3.4. **Obstruction to finer filtered morphisms.** Let

$$\widetilde{\beta}_s : F^Y_\bullet K_Y \longrightarrow F^X_{\bullet+s} K_X$$

be a filtered $\mathcal{A}$-linear lift of the fixed comparison β. We improve its degree while retaining the filtered module structure and the identification of its underlying morphism with β. Put $\Delta = d_Y - d_X$.

Lemma 3.15. *The obstruction to factoring $\widetilde{\beta}_s$ through the transition morphism*

$$F^X_{\bullet+s-1} K_X \longrightarrow F^X_{\bullet+s} K_X$$

in the category of filtered $\mathcal{A}$-modules is a class in

$$\mathrm{Ext}^{-s}_{E,\Delta+s}(U, V).$$

If this group vanishes, there exists a filtered $\mathcal{A}$-linear lift of β of degree $s-1$ whose composite with the transition morphism is $\widetilde{\beta}_s$.

Proof. Form the Rees algebra and modules

$$R = \bigoplus_n F_n\mathcal{A}\, t^n, M = \bigoplus_n a^*({}^p\tau_{\le n} K_Y) t^n, N = \bigoplus_n ({}^p\tau_{\le n} K_X) t^n.$$

Multiplication by t is the filtration transition map. The Rees construction represents the filtered lift by a morphism

$$\Phi_s \in \mathrm{Hom}_{D_{\mathrm{gr}}(R)}(M, N\langle s\rangle) = H^0 R\Gamma\big(B_X, R\mathcal{H}om^{\mathrm{gr}}_R(M, N\langle s\rangle)\big).$$

Here R is an internally graded algebra object; its cohomological and internal degrees are distinct. All quotients below are derived; in particular, $\bar{R} = R \otimes^{\mathbf{L}}_{\mathbb{Q}_\ell[t]} \mathbb{Q}_\ell$, and likewise $\bar{M} = M/t$ and $\bar{N} = N/t$, with t acting by zero on $\mathbb{Q}_\ell$. Then

$$\bar{R}_{-k} \simeq \Lambda^k T(Q)[k], \qquad \bar{M}_n \simeq a^*({}^p\mathcal{H}^n K_Y)[-n], \qquad \bar{N}_n \simeq {}^p\mathcal{H}^n K_X[-n].$$

Their sheared algebra and modules are precisely E, U, V defined above. After forgetting the grading, specialization at $t = 1$ gives

$$M/(t-1) \simeq a^* K_Y, \qquad N/(t-1) \simeq K_X, \qquad \Phi_s|_{t=1} = \beta.$$

The triangle in $D_{\mathrm{gr}}(R)$

$$N\langle s-1\rangle \xrightarrow{t} N\langle s\rangle \xrightarrow{q_s} \bar{N}\langle s\rangle \xrightarrow{+1}$$

shows that the obstruction to lowering s is $o(\Phi_s) = [q_s \Phi_s]$. Derived extension of scalars and shearing give

$$R\mathcal{H}om^{\mathrm{gr}}_R(M, \bar{N}\langle s\rangle) \simeq R\mathcal{H}om^{\mathrm{gr}}_{\bar{R}}(\bar{M}, \bar{N}\langle s\rangle) \simeq R\mathcal{H}om^{\mathrm{gr}}_E(U, V\langle \Delta + s\rangle[-s]).$$

Indeed, writing $n = d_Y + i$, a component of the middle expression is

$$a^*({}^p\mathcal{H}^n K_Y)[-n] \longrightarrow {}^p\mathcal{H}^{n+s} K_X[-n-s], \qquad n + s = d_X + (i + \Delta + s),$$

which becomes $U_i \to V_{i+\Delta+s}[-s]$ after shifting by $[n]$. Taking $H^0 R\Gamma(B_X, -)$ therefore places $o(\beta_s)$ in $\mathrm{Ext}^{-s}_{E,\Delta+s}(U, V)$. If this group vanishes, the long exact Hom sequence gives

$$\Phi_{s-1} : M \longrightarrow N\langle s-1\rangle, \qquad t\Phi_{s-1} = \Phi_s.$$

Under the Rees correspondence, this gives the required filtered module morphism of degree $s-1$. Specialization at $t = 1$ shows that its underlying comparison is still β. □

3.5. **Conclusion of the main argument.**

Lemma 3.16. *Under the geometric hypotheses, suppose that, for some $s \in \mathbb{Z}$ and every $n \in \mathbb{Z}$, the natural comparison with $\mathbb{Q}_\ell$-coefficients induces factorizations*

$$a^*({}^p\tau_{\le n}K_{Y,\mathbb{Q}_\ell}) \longrightarrow {}^p\tau_{\le n+s}K_{X,\mathbb{Q}_\ell} \longrightarrow K_{X,\mathbb{Q}_\ell}.$$

Then the corresponding factorizations exist over $\mathbb{Q}$. They are unique in the ordinary derived category and commute with the truncation transition maps.

Proof. This follows from the standard comparison argument of [BBD82, §6.1] and faithful flatness of $\mathbb{Q}_\ell/\mathbb{Q}$. □

Proof of Theorem 3.5. On a fixed stratum S of dimension t on which $r(b)$ is a constant r, the stratum is contained in $\{b : r(b) \le r\}$. The rank hypothesis therefore gives $t \le r$. Lemmas 3.6–3.8 give the natural $\mathcal{A}$-module morphism β and a filtered $\mathcal{A}$-linear lift

$$\widetilde{\beta}_{d_X} : F^Y_\bullet K_Y \longrightarrow F^X_{\bullet+d_X} K_X.$$

Lemma 3.10 and Proposition 3.11 provide the geometric inputs to Lemma 3.13; Proposition 3.14 then gives $\mathrm{Ext}^e_{E,\delta}(U,V) = 0$ for $e < \Delta + \delta$.

For $s > -\Delta$, put $\delta = \Delta + s > 0$. The obstruction in Lemma 3.15 vanishes because $-s = \Delta - \delta < \Delta + \delta$.

Starting at $s = d_X$, we lower the degree finitely many times to $s = -\Delta = d_X - d_Y$. Forgetting the module structure and passing to the homotopy category gives factorizations

$$a^*({}^p\tau_{\le n}K_{Y,\mathbb{Q}_\ell}) \longrightarrow {}^p\tau_{\le n+d_X-d_Y}K_{X,\mathbb{Q}_\ell} \longrightarrow K_{X,\mathbb{Q}_\ell}$$

of the natural comparison for every n.

Lemma 3.16 gives the corresponding factorizations over $\mathbb{Q}$. Setting $n = d_Y + k$ proves (3.1), and hence compatibility in the sense of Definition 2.2. The assertion about the perverse filtration follows by hypercohomology. □

Keep the geometric hypotheses preceding Theorem 3.5, but replace its rank inequality by

$$\dim\{b \in B_X : r(b) \le c\} \le c + \kappa \qquad (c \ge 0), \qquad \kappa \in \mathbb{Z}_{\ge 0}. \tag{3.5}$$

Corollary 3.17. *The natural comparison morphism admits compatible factorizations*

$$a^{*p}\tau_{\le d_Y+k}K_Y \longrightarrow {}^p\tau_{\le d_X+k+\kappa}K_X \qquad (k \in \mathbb{Z}). \tag{3.6}$$

In particular, $F^*P_kH^m(Y,\mathbb{Q}) \subseteq P_{k+\kappa}H^m(X,\mathbb{Q})$.

Proof. On a fixed stratum S, $r \ge t - \kappa$, so the proof of Proposition 3.14 gives

$$\mathrm{Ext}^e_{E,\delta}(U,V) = 0 \qquad (e < \Delta + \delta - 2\kappa).$$

If $\kappa \ge d_Y$, Lemma 3.8, followed by the canonical truncation transition maps, gives the factorizations over $\mathbb{Q}_\ell$. They descend to $\mathbb{Q}$ by Lemma 3.16.

Assume now that $0 \le \kappa < d_Y$. For $s > \kappa - \Delta$, we have

$$-s < \Delta + (\Delta + s) - 2\kappa.$$

Lemma 3.15 therefore successively improves the filtered degree to $\kappa - \Delta$, giving a filtered $\mathcal{A}$-linear lift

$$\widetilde{\beta}_{\kappa-\Delta} : F^Y_\bullet K_Y \longrightarrow F^X_{\bullet+\kappa-\Delta} K_X.$$

Forgetting the module structure and passing to the homotopy category gives the corresponding truncation factorizations over $\mathbb{Q}_\ell$. Lemma 3.16 descends these factorizations to $\mathbb{Q}$. Writing $n = d_Y + k$ proves (3.6). □

4. Multiplicative perverse truncations for Hitchin Systems

4.1. **The strongly parabolic Hitchin systems.** Let G be a connected reductive group, C a smooth connected projective curve, and $D \subset C$ a reduced divisor, possibly empty. Put $L = K_C(D)$. Fix a Borel subgroup $\mathsf{B} \subset G$ and put $\mathfrak{b} = \operatorname{Lie}\mathsf{B}$, $\mathfrak{n} = \operatorname{Lie} R_u(\mathsf{B})$, where $R_u(\mathsf{B})$ is the unipotent radical. Let $\mathcal{G}$ be the Iwahori group scheme on C determined by

$$\mathcal{G}|_{C\setminus D} = G \times (C \setminus D), \qquad \mathcal{G}(\widehat{\mathcal{O}}_{C,p}) = \{g \in G(\widehat{\mathcal{O}}_{C,p}) : g(p) \in \mathsf{B}\} \quad (p \in D).$$

For $D = \varnothing$, take $\mathcal{G} = G \times C$. Let $\mathrm{Bun}_{\mathcal{G}}$ be the moduli stack of $\mathcal{G}$-bundles. For a scheme S, an object of $\mathrm{Bun}_{\mathcal{G}}(S)$ can be interpreted as a principal G-bundle E on $C \times S$ together with sections $\sigma_p : S \longrightarrow (E/\mathsf{B})|_{\{p\}\times S}$, $p \in D$, see [Hei10, Section 1, Example (2)] and [MRR23, Theorem 4.8 and Example 4.9]. Morphisms are bundle isomorphisms preserving these sections. Set $\operatorname{ad} E := E \times^G \mathfrak{g}$, where $\mathfrak{g} = \operatorname{Lie} G$ and G acts on $\mathfrak{g}$ by the adjoint representation. Put $E_p := E|_{\{p\}\times S}$. Pulling back the B-torsor $E_p \to E_p/\mathsf{B}$ along the section $\sigma_p : S \to E_p/\mathsf{B}$ gives

$$E_{\mathsf{B},p} := S \times_{\sigma_p,\, E_p/\mathsf{B}} E_p.$$

For a geometric object (E, σ), define

$$\mathfrak{b}_p := E_{\mathsf{B},p} \times^{\mathsf{B}} \mathfrak{b} \subset (\operatorname{ad} E)_p, \qquad \mathfrak{n}_p := E_{\mathsf{B},p} \times^{\mathsf{B}} \mathfrak{n} \subset \mathfrak{b}_p,$$

where both contracted products use the adjoint action of B. With $i_p : \{p\} \hookrightarrow C$, set

$$\mathrm{ParAd}(E) = \ker\left(\operatorname{ad} E \longrightarrow \bigoplus_{p\in D} i_{p*}((\operatorname{ad} E)_p/\mathfrak{b}_p)\right),$$

$$\mathrm{SParAd}(E) = \ker\left(\operatorname{ad} E \longrightarrow \bigoplus_{p\in D} i_{p*}((\operatorname{ad} E)_p/\mathfrak{n}_p)\right).$$

For the $\mathcal{G}$-torsor $\mathcal{E}$ corresponding to (E, σ), we can also define the adjoint bundle using the Lie algebra of the group scheme $\mathcal{G}$. In fact, $\operatorname{ad}(\mathcal{E}) = \mathrm{ParAd}(E)$.

Definition 4.1. A *strongly parabolic Higgs bundle* is a triple (E, σ, ϕ) with

$$\phi \in H^0(C, \mathrm{SParAd}(E) \otimes L), \qquad \text{equivalently} \qquad \operatorname{Res}_p \phi \in \mathfrak{n}_p \quad (p \in D). \tag{4.1}$$

For a scheme S, let $\mathcal{M}_{\mathcal{G}}(S)$ be the groupoid of triples (E, σ, ϕ), where (E, σ) is an object of $\mathrm{Bun}_{\mathcal{G}}(S)$ and

$$\phi \in H^0(C \times S, \operatorname{ad} E \otimes \mathrm{pr}_C^* L), \qquad \operatorname{Res}_p \phi \in H^0(S, E_{\mathsf{B},p} \times^{\mathsf{B}} \mathfrak{n}) \quad (p \in D).$$

Here $\mathrm{pr}_C : C \times S \to C$ is the projection, and Res_p denotes the relative residue along $\{p\} \times S$. Morphisms are isomorphisms of G-bundles preserving the sections and the Higgs fields. For

a morphism $f : S' \to S$, pullback is obtained by pulling back E and ϕ along $\mathrm{id}_C \times f$ and each flag section σ_p along f.

Choose a nondegenerate invariant symmetric form κ on $\mathfrak{g}$. Since $\mathfrak{b}_p^{\perp} = \mathfrak{n}_p$,

$$\mathrm{ParAd}(E)^{\vee} \otimes K_C \simeq \mathrm{SParAd}(E) \otimes L, \qquad \mathcal{M}_{\mathcal{G}} \simeq T^* \operatorname{Bun}_{\mathcal{G}}, \tag{4.2}$$

where $T^* \operatorname{Bun}_{\mathcal{G}}$ denotes the cotangent stack of $\operatorname{Bun}_{\mathcal{G}}$. Let $\mathfrak{c}_G = \mathfrak{g}/\!/G$. Choose homogeneous generators of $\mathbb{C}[\mathfrak{g}]^G$ of degrees $d_1, \dots, d_{\operatorname{rk} G}$. The Hitchin base is

$$B_G(C, D) = \bigoplus_{i=1}^{\operatorname{rk} G} H^0\big(C, L^{d_i}(-D)\big). \tag{4.3}$$

For $D = \varnothing$, we call it *the canonical Hitchin base*. Denote the chosen invariant generators by f_i. The Hitchin morphism is

$$h(E, \sigma, \phi) = (f_i(\phi))_i \in B_G(C, D), \qquad f_i(\phi)|_D = f_i(\operatorname{Res}_D \phi) = 0.$$

Thus its target is (4.3). For $D = \varnothing$, the definitions give $\mathrm{ParAd}(E) = \mathrm{SParAd}(E) = \operatorname{ad} E$. For more details, see [Yun11].

Fix the class $\tau \in \pi_1(G)$ of the underlying topological bundle of E. Choose a maximal torus $T \subset \mathsf{B}$, with cocharacter lattice $X_*(T) := \operatorname{Hom}(\mathbb{G}_m, T)$. Let $\mathfrak{A} \subset X_*(T) \otimes_{\mathbb{Z}} \mathbb{R}$ be the open Weyl alcove whose closure contains 0 and whose associated parabolic subgroup is B, using the conventions of [BGPMiR20]. Fix rational parabolic weights $\boldsymbol{\alpha} = (\alpha_p)_{p \in D}$ with

$$\alpha_p \in \mathfrak{A} \cap \big(X_*(T) \otimes_{\mathbb{Z}} \mathbb{Q}\big) \qquad (p \in D).$$

For $D = \varnothing$ the weight tuple is empty. Parabolic degree and (semi)stability are defined by [BGPMiR20], using holomorphic reductions of the flagged bundle above and the central parameter fixed by its parabolic degrees. We denote the $\boldsymbol{\alpha}$-semistable and stable loci of type τ by

$$\mathcal{M}_{\mathcal{G}}^{\tau, ss}(\boldsymbol{\alpha}), \qquad \mathcal{M}_{\mathcal{G}}^{\tau, s}(\boldsymbol{\alpha}) \subset \mathcal{M}_{\mathcal{G}}.$$

We write $M_G(C, D)$ for the coarse moduli space of the semistable locus, with the fixed parameters τ and $\boldsymbol{\alpha}$ omitted from the notation.

4.1.1. *Deformations and symplectic forms.* For $x = (E, \sigma, \phi)$, put

$$\mathcal{C}_x^{\bullet} = \Big[\mathrm{ParAd}(E) \xrightarrow{[\phi, -]} \mathrm{SParAd}(E) \otimes L\Big], \qquad \deg \mathcal{C}_x^{\bullet} = 0, 1.$$

The deformation calculation of [BKV19, Section 4.1] gives

$$\begin{gathered} \operatorname{Lie} \operatorname{Aut}(x) = \mathbb{H}^0(C, \mathcal{C}_x^{\bullet}), \qquad T_x \mathcal{M}_{\mathcal{G}} = \mathbb{H}^1(C, \mathcal{C}_x^{\bullet}), \\ \mathbb{H}^i(C, \mathcal{C}_x^{\bullet}) \simeq \mathbb{H}^{2-i}(C, \mathcal{C}_x^{\bullet})^{\vee}. \end{gathered} \tag{4.4}$$

The group $\mathbb{H}^2(C, \mathcal{C}_x^{\bullet})$ is the obstruction space for extending a deformation of $x = (E, \sigma, \phi)$ from an Artinian base to an infinitesimal thickening of that base, with (C, D) fixed. Set

$$\mathcal{C}_{x,\mathrm{cen}}^{\bullet} := [\mathfrak{z}_G \otimes \mathcal{O}_C \xrightarrow{0} \mathfrak{z}_G \otimes K_C].$$

Let $\mathcal{C}_{x,\mathrm{der}}^{\bullet} := \ker(\mathcal{C}_x^{\bullet} \to \mathcal{C}_{x,\mathrm{cen}}^{\bullet})$, where the map is induced by the projection onto $\mathfrak{z}_G$ along $[\mathfrak{g}, \mathfrak{g}]$. If $\operatorname{Lie} \operatorname{Aut}(x) = \mathfrak{z}_G$, then

$$\mathbb{H}^0(C, \mathcal{C}_{x,\mathrm{der}}^{\bullet}) = \mathbb{H}^2(C, \mathcal{C}_{x,\mathrm{der}}^{\bullet}) = 0.$$

Since $[\mathfrak{z}_G, \mathfrak{g}] = 0$, the above vanishing and [Man04, Propositions 5.47 and 5.49] imply that $\mathcal{M}_{\mathcal{G}}$ is smooth at x.

Let Z_G^0 be the identity component of the center of G, with Lie algebra $\mathfrak{z}_G$, and let $r : \mathcal{M}_{\mathcal{G}}^{\tau,s}(\boldsymbol{\alpha}) \to \mathfrak{X}$ be its rigidification, in the sense of [AOV08, Appendix A, Theorem A.1]:

$$\mathrm{Aut}_{\mathfrak{X}}(r(x)) = \mathrm{Aut}(x)/Z_G^0.$$

In other words, $r : \mathcal{M}_{\mathcal{G}}^{\tau,s}(\boldsymbol{\alpha}) \to \mathfrak{X}$ is a gerbe banded by Z_G^0. Hence, $\mathfrak{X}$ is smooth. Since the automorphism groups have zero Lie algebra, $\mathfrak{X}$ is a Deligne–Mumford stack.

Under (4.2), let λ_κ be the Liouville one-form on $\mathcal{M}_{\mathcal{G}}$, and set $\omega_\kappa = d\lambda_\kappa$. The restriction of ω_κ to the stable locus descends to a closed nondegenerate two-form on $\mathfrak{X}$; see [BGL15, Sections 2 and 3.1–3.2] and [BKV19, Section 4.1, equation (7)]. We omit the subscript κ when the form is fixed.

4.2. **Regular centralizers and Picard actions.** Let $\chi : \mathfrak{g} \to \mathfrak{c}_G$ be the adjoint quotient. Ngô constructs a smooth commutative regular-centralizer group scheme $J_G \to \mathfrak{c}_G$ and a homomorphism $\chi^* J_G \to I_G$ to the universal automorphism centralizer on $\mathfrak{g}$ [Ngô10, Section 2.1]. Twisting by the Higgs coefficient line bundle and pulling back along a characteristic b gives a group scheme $J_b \to C$. For Borel reductions, Yun constructs the compatible homomorphism to the flag-preserving centralizer and the corresponding Picard action [Yun11, Lemmas 2.3.1–2.3.3]. Write $B = B_G(C, D)$ and let $J \to C \times B$ be the regular centralizer above. The relative Picard stack is, by definition,

$$\mathcal{P}(S) = \{J|_{C\times S}\text{-torsors on } C \times S\}, \qquad S \longrightarrow B,$$

with torsor isomorphisms as morphisms and contracted product as the group law. It is smooth over B [dCFFHM25, Definition 4.13 and Proposition 4.16]. For $x = (E, \phi, (E_{\mathsf{B},p})_p)$ and $b = h(x)$, by [Yun11, Lemmas 2.3.1–2.3.3], the centralizer homomorphisms j_G, j_{B} give

$$\mathcal{T} \cdot x = \left(\mathcal{T} \times^{J_b, j_G} (E, \phi), (\mathcal{T}|_p \times^{J_b|_p, j_{\mathsf{B}}} E_{\mathsf{B},p})_p\right). \tag{4.5}$$

Here $E_{\mathsf{B},p}$ is the Borel torsor defined by σ_p, $j_G : J_b \to \underline{\mathrm{Aut}}_C(E, \phi)$ is the centralizer homomorphism, and j_{B} is its flag-preserving factorization at p. That action preserves $\mathfrak{b}_p \to \mathfrak{b}_p/\mathfrak{n}_p$; its zero fiber is exactly the residue condition (4.1).

Let $P = (\pi_0^{\mathrm{fppf}}\mathcal{P})^0$ whenever this identity component is represented by a smooth group scheme; here $\pi_0^{\mathrm{fppf}}\mathcal{P}$ is the fppf sheafification of the presheaf of isomorphism classes. Set $\mathcal{P}^0 = \mathcal{P} \times_{\pi_0^{\mathrm{fppf}}\mathcal{P}} P$ and write $\rho : \mathcal{P}^0 \to P$ for the projection. Under the finite-inertia and component-preservation hypotheses in Assumption 4.2(1)–(2), the torsor action (4.5) induces the compatible actions

$$\begin{array}{ccc} \mathcal{P}^0 \times_B \mathfrak{X} & \xrightarrow{\widetilde{c}} & \mathfrak{X} \\ {\scriptstyle \rho\times q}\downarrow & & \downarrow{\scriptstyle q} \\ P \times_B X & \xrightarrow[c]{} & X, \end{array} \tag{4.6}$$

where $q : \mathfrak{X} \to X$ is the coarse moduli morphism [AOV08, Section 3]. The upper action is by $\mathcal{P}^0$. An action of P itself on $\mathfrak{X}$ would require further descent: for example, [dCFFHM25, Corollary 5.35] rigidifies by the whole Z_G and assumes $\deg L > 0$.

4.2.1. *The Hamiltonian identity.* Set $\mathcal{W} = \bigoplus_i L^{d_i}(-D)$, so $B = H^0(C, \mathcal{W})$. The regular-centralizer identity [dCFFHM25, Proposition 4.12(iii)] and Serre duality give

$$\begin{aligned} \operatorname{Lie} J_b &\simeq \bigoplus_i L^{1-d_i} = \mathcal{W}^\vee \otimes K_C, \\ \eta_b : H^1(C, \operatorname{Lie} J_b) &\xrightarrow{\sim} H^0(C, \mathcal{W})^\vee = T_b^* B. \end{aligned} \tag{4.7}$$

Differentiating j_G, j_{B} and the invariant polynomials gives maps of complexes

$$\operatorname{Lie} J_b[0] \xrightarrow{(dj_\phi, 0)} \mathcal{C}_x^\bullet \xrightarrow{(0, d\chi_\phi)} \mathcal{W}[-1].$$

The two maps of complexes are dual under the pairing induced by κ, as expressed by the local identity

$$\kappa(dj_\phi(s), v) = \langle s, d\chi_\phi(v) \rangle \quad (s \in \operatorname{Lie} J_b,\ v \in \operatorname{SParAd}(E) \otimes L);$$

see [dCHM21, Lemma A.9 and Remark A.11] and [dCFFHM25, Lemma 7.11]. For the corresponding global duality in the unmarked K_C-valued case, see [dCHM21, Proposition A.12].

For $\xi \in H^1(C, \operatorname{Lie} J_b)$, let $\xi^\# \in T_x\mathfrak{X}$ be the tangent vector obtained by acting on x with the first-order deformation of the trivial J_b-torsor represented by ξ. Taking first hypercohomology of the two maps above gives

$$H^1(C, \operatorname{Lie} J_b) \xrightarrow{\xi \mapsto \xi^\#} T_x\mathfrak{X} \xrightarrow{dh_x} T_b B = H^0(C, \mathcal{W}),$$

where dh_x is the differential of the Hitchin map. Hence

$$\omega_x(\xi^\#, w) = -\langle \eta_b(\xi), dh_x(w) \rangle, \qquad \xi \in H^1(C, \operatorname{Lie} J_b), \quad w \in T_x\mathfrak{X}. \tag{4.8}$$

Let $p : C \times B \to B$ be the projection. When the differential of ρ induces an isomorphism $\theta : R^1 p_*(\operatorname{Lie} J) \xrightarrow{\sim} \operatorname{Lie}(P/B)$, (4.7) gives $\eta : \operatorname{Lie}(P/B) \xrightarrow{\sim} \Omega_B^1$ via θ^{-1}. Then (4.8) reads $\iota_{\xi^\#}\omega = -h^*\eta(\xi)$ for local sections ξ of $\operatorname{Lie}(P/B)$.

4.3. **Geometric hypotheses for the applications.** The following conditions are used in multiplicativity and in Theorem 1.1. They concern the complete semistable moduli space in the chosen topological type and with the specified weights.

Assumption 4.2. *Let X be a chosen Higgs moduli space with Hitchin base $B = B_G(C, D)$, and put $d = \dim B$. Assume the following conditions are satisfied:*

(1) *The chosen component satisfies*

$$\varnothing \neq \mathcal{M}_{\mathcal{G}}^{\tau, ss}(\boldsymbol{\alpha}) = \mathcal{M}_{\mathcal{G}}^{\tau, s}(\boldsymbol{\alpha}).$$

The rigidified stack $\mathfrak{X}$ above has finite inertia, meaning that

$$I_{\mathfrak{X}} := \mathfrak{X} \times_{\mathfrak{X} \times \mathfrak{X}} \mathfrak{X} \longrightarrow \mathfrak{X} \quad \text{is finite.}$$

Its coarse moduli space is X.

(2) *The action of $\mathcal{P}^0$ by twisting with J_b-torsors preserves the chosen stable component $\mathcal{M}_{\mathcal{G}}^{\tau, s}(\boldsymbol{\alpha})$, (4.6) holds, and the natural map induced by ρ is an isomorphism:*

$$\theta : R^1 p_*(\operatorname{Lie} J) \xrightarrow{\sim} \operatorname{Lie}(P/B).$$

Moreover (X, P, B) is a weak abelian fibration in the sense of Definition 2.4.

(3) *The Hitchin map $h_X : X \to B$ is projective and surjective, and*

$$\dim X = 2d, \qquad X_b \text{ is connected and pure of dimension } d \quad (b \in B).$$

Moreover,

$$\dim\{b \in B : \dim \mathrm{Ab}(P_b) \le c\} \le c \qquad (c \ge 0).$$

Under Assumption 4.2, the preceding deformation calculation shows that $\mathfrak{X}$ is smooth. Its finite inertia implies that X has finite quotient singularities, hence is a rational homology manifold.

Assumption 4.3. *For a comparison $(F, a) : (X \to B_X) \to (Y \to B_Y)$, there is a homomorphism $\varphi : P_X \to a^* P_Y$ over B_X such that*

$$F_a(p \cdot x) = \varphi(p) \cdot F_a(x), \qquad F_a(x) = (F(x), h_X(x)).$$

Under this assumption, we take $Q = P_X$, $u = \mathrm{id}$ and $v = \varphi$ in Theorem 3.5. The effective-rank inequality is then the rank inequality in Assumption 4.2(3). For the diagonal, equivariance holds with $\varphi = \Delta_P$, as used in Theorem 4.4 below.

4.4. **Proof of multiplicativity.** In this subsection, we prove the multiplicativity for the perverse truncation of direct image sheaves.

The product $h \times h : X \times X \to B \times B$ is the Hitchin fibration for $G \times G$, with the prescribed moduli data on each factor. Indeed,

$$\mathbb{C}[\mathfrak{g} \oplus \mathfrak{g}]^{G \times G} = \mathbb{C}[\mathfrak{g}]^G \otimes \mathbb{C}[\mathfrak{g}]^G;$$

see the construction in [dCFFHM25, Definitions 3.7–3.8]. The diagonal homomorphism $G \to G \times G$ induces Δ_X on these moduli spaces and Δ_B on their Hitchin bases, with

$$(h \times h) \circ \Delta_X = \Delta_B \circ h.$$

Theorem 4.4. *Let $h : X \to B$ satisfy Assumption 4.2, and put $K = Rh_* \mathbb{Q}_X$ and $d = \dim B$. Then, for $r, s \in \mathbb{Z}$, the cup product factors as*

$$ {}^p\tau_{\le r} K \otimes^{\mathbf{L}} {}^p\tau_{\le s} K \longrightarrow {}^p\tau_{\le r+s-d} K \longrightarrow K. \tag{4.9}$$

In particular $P_i H^m(X, \mathbb{Q}) \cup P_j H^n(X, \mathbb{Q}) \subseteq P_{i+j} H^{m+n}(X, \mathbb{Q})$.

Proof. Consider the diagonal maps

$$F = \Delta_X : X \to X \times X, \qquad a = \Delta_B : B \to B \times B,$$

which satisfy $(h \times h) \circ F = a \circ h$. Take $Q = P$, with homomorphisms

$$P \xleftarrow{\mathrm{id}} Q = P \xrightarrow{\Delta_P} \Delta_B^*(P \times P).$$

Both homomorphisms have trivial kernel, so the effective rank at $b \in B$ is $\dim \mathrm{Ab}(P_b)$. The map $h \times h$ is projective, its total space has dimension $4d$, and its fibers are connected and pure of dimension $2d$. Moreover, $\mathrm{IC}_{X \times X} \simeq \mathrm{IC}_X \boxtimes \mathrm{IC}_X$, so $X \times X$ is a rational homology manifold. Since

$$ {}^p\tau_{\le r} K \boxtimes {}^p\tau_{\le s} K \in {}^p D^{\le r+s}(B \times B),$$

Theorem 3.5, with $k = r + s - 2d$, gives

$$\begin{aligned} {}^{p}\tau_{\le r}K \otimes^{\mathbf{L}} {}^{p}\tau_{\le s}K &\simeq \Delta_B^*\big({}^{p}\tau_{\le r}K \boxtimes {}^{p}\tau_{\le s}K\big) \\ &\longrightarrow \Delta_B^*\,{}^{p}\tau_{\le r+s}(K \boxtimes K) \longrightarrow {}^{p}\tau_{\le r+s-d}K \longrightarrow K. \end{aligned}$$

Under proper Künneth, pullback along Δ_X identifies this composite with the cup product on K. Taking $r = d + i$, $s = d + j$ and hypercohomology proves the assertion for $P_\bullet$ by (2.2). □

Remark 4.5. Iterating (4.9) gives, for every integer $\ell \ge 2$ and $r_1, \dots, r_\ell \in \mathbb{Z}$, factorizations

$$\bigotimes_{i=1}^{\ell} {}^{p}\tau_{\le r_i}K \longrightarrow {}^{p}\tau_{\le \sum_{i=1}^{\ell} r_i - (\ell-1)d}K \longrightarrow K$$

whose composite is the ℓ-fold cup-product.

Replace the rank inequality in Assumption 4.2(3) by $\dim\{b \in B : \dim \mathrm{Ab}(P_b) \le c\} \le c+\kappa$ for all $c \ge 0$, with $\kappa \in \mathbb{Z}_{\ge 0}$, and keep the other hypotheses.

Corollary 4.6. *For every $r, s \in \mathbb{Z}$, the cup product on $K = Rh_*\mathbb{Q}_X$ factors as*

$$ {}^{p}\tau_{\le r}K \otimes^{\mathbf{L}} {}^{p}\tau_{\le s}K \longrightarrow {}^{p}\tau_{\le r+s-d+\kappa}K \longrightarrow K.$$

Consequently $P_iH^m(X,\mathbb{Q}) \cup P_jH^n(X,\mathbb{Q}) \subseteq P_{i+j+\kappa}H^{m+n}(X,\mathbb{Q})$.

Proof. Repeat the arguments in Theorem 4.4 with Corollary 3.17 at $k = r + s - 2d$. □

Let C be a smooth projective curve of genus at least two, and fix $N \ge 2$, $e \in \mathbb{Z}$ and $\Lambda \in \mathrm{Pic}^e(C)$ with $\gcd(N, e) = 1$. Let $M_{N,e}$ be the moduli space of semistable rank-N, degree-e, K_C-valued Higgs bundles, and let $M_{\mathrm{SL}_N}^{\Lambda,0}(C, \varnothing)$ denote its closed subspace defined by $\det E \simeq \Lambda$ and $\operatorname{tr}\phi = 0$.

Corollary 4.7. *Let $X = M_{N,e}$ or $X = M_{\mathrm{SL}_N}^{\Lambda,0}(C, \varnothing)$, and let $h : X \to B$ be its Hitchin morphism. Put $K = Rh_*\mathbb{Q}_X$ and $d = \dim B$. Then cup-product satisfies (4.9) for all $r, s \in \mathbb{Z}$. In particular,*

$$P_iH^a(X,\mathbb{Q}) \cup P_jH^b(X,\mathbb{Q}) \subseteq P_{i+j}H^{a+b}(X,\mathbb{Q}).$$

Proof. The geometric hypotheses follow from [dCHM21, Section 2.2] and [dCMS22, Section 1.1]. For the fixed-determinant space, the spectral norm identifies the acting neutral Prym group with the SL_N regular-centralizer Picard group; preservation of stability and affine stabilizers are inherited from the GL_N action. The required group properties, Lie identification and full-base δ-regularity follow from [dCFFHM25, Propositions 4.12, 4.17, 4.29, Corollary 4.27 and Theorem 7.16]. The rational Betti polarization is supplied by [AF24, Theorem 1.2]. Apply Theorem 4.4. □

In particular, the cohomological consequence gives another proof of multiplicativity of the perverse filtration for coprime GL_N Hitchin moduli [MS24, HMMS25, MSY25] and fixed-determinant traceless SL_N Hitchin moduli [Wen26, Theorem 1.1]. For related motivic and sheaf-theoretic multiplicativity results of Maulik–Shen–Yin, see [MSY25, Theorem 2.6(ii)–(iii)], [MSY26a], and [MSY26b, Theorem 3.2(b)].

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